DISTANCE Chapter 05

The Spectrum of Dynamicity

Kangbin Yim*

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1 A Boundary That Appears Absolute

Few distinctions seem as fundamental to us as the distinction between life and death. A tree grows, extends its branches toward light, absorbs water through its roots, and produces new leaves. An animal moves, searches for food, avoids danger, and responds to its surroundings. A human being speaks, remembers, anticipates the future, and changes the surrounding environment according to intention. A stone, by contrast, displays none of these familiar signs. A dead organism no longer breathes, responds to stimuli, or repairs itself, and the countless activities that once allowed us to recognize it as a single living being are no longer sustained. The difference seems unmistakable. Life appears either to be present or absent; some things are regarded as alive, while those that are not alive are placed on the other side of the divide, and death is accepted as the definite boundary separating the two states.

This distinction is so deeply embedded in our language and thought that we rarely ask why it exists in precisely this form. Living organisms are treated as belonging to a fundamentally different domain from nonliving things, biology begins by identifying the characteristics of life, and death is understood as the point at which those characteristics disappear. Such distinctions are, of course, extremely useful. Medicine must determine whether an organism still maintains integrated physiological functions. Ecology must distinguish living populations from the physical environments surrounding them. Law, ethics, and social institutions likewise require criteria by which birth, life, and death can be recognized. Yet the usefulness of a distinction does not necessarily mean that it

*email: yim@sch.ac.kr (Prof. SCH University, Korea)

corresponds to the most fundamental distinction in the Universe. The boundary between life and death may appear so clear not because the boundary itself is absolute, but because of the scale and position from which we observe it.

When we look at an organism as a whole, death can seem remarkably abrupt. Breathing stops, blood circulation ceases, and the integrated activity of the nervous system collapses until the organism can no longer respond to its surroundings as a single individual. Yet not every structure that constituted the body disappears at the same moment. Cells continue to function for different lengths of time. Various chemical reactions persist, microorganisms begin to proliferate, molecules form new combinations, heat disperses into the surroundings, and matter passes into the soil, the atmosphere, and eventually other living organisms. What has disappeared is not activity itself. What has disappeared is a particular mode of organization that once bound countless activities together as one living organism.

The same difficulty appears when we look in the opposite direction and ask when life begins. A seed may remain dormant for years while displaying almost no visible biological activity. A virus can appear almost chemically inert outside a host, yet become extraordinarily active once it enters a living cell. Some spores remain at extremely low levels of activity for such long periods that it becomes difficult to decide whether they should be described as alive or merely preserved. Where, then, does life begin? The question is far less easily answered than our ordinary distinction between life and nonlife might suggest.

Other examples disturb our intuition even further. Fire consumes surrounding material, spreads, transforms its environment, and under suitable conditions continuously reproduces its form, yet we do not call fire a living organism. A crystal incorporates material from its surroundings, grows, and develops a regular structure, but it too is not classified as alive. A planet maintains enormous internal and external cycles while preserving its overall structure over immense periods. A star is born, sustains its structure for a long time, transforms matter, releases enormous amounts of energy, and eventually loses the configuration by which we once recognized it. Yet we do not ordinarily describe planets or stars as living.

None of this means that stones, fires, crystals, planets, and living organisms are all the same. Rather, these examples show that the boundary between life and nonlife cannot be explained simply by the presence or absence of change. Nothing in the Universe is entirely without change. Nor can the distinction be reduced to the presence or absence of motion.

Rivers flow, storms continually transform their own structures, and planets rotate and orbit. Even within a stone that appears motionless, atoms and molecules remain involved in continuing motion and interaction. The boundary between life and nonlife therefore cannot be established merely by asking whether something moves.

The difficulty may not arise only because our definition of life remains incomplete. Another assumption may be hidden in the way we frame the question itself. DISTANCE does not begin by dividing life and nonlife into two fundamentally different kinds of existence. Instead, everything we recognize as a distinct entity can be understood as an **Island**: a configuration that maintains an effective boundary for some duration and is thereby distinguishable from its surroundings. An Island may be formed through the combination of smaller Islands and need not be completely isolated from what lies beyond it. A stone is an Island. A tree is an Island. A cell, an animal, a planet, and a star can each be regarded as an Island. Each maintains internal relationships while simultaneously remaining involved in continuing relationships with what surrounds it, sustaining a distinguishable configuration while being affected by Momentum that becomes effective both within and beyond its boundary. No Island stands outside the continuing reconfiguration of the Universe.

A stone lying in a field appears almost unchanged, but this appearance depends heavily on the scale at which humans observe it. The stone exchanges heat with its surroundings, experiences pressure and chemical alteration, develops fractures, weathers, and is eventually incorporated into other configurations. Its visible transformation may be slow, but it is not without change. A star presents a very different appearance. Within it, enormous energy differences are continuously being resolved, while Momentum becomes effective on a vast scale, with consequences extending far beyond the star's internal configuration to countless surrounding Islands. A living organism exhibits still another level of complexity. It takes in external structures, breaks them down internally, reorganizes them into new structures, and releases other structures back into its surroundings. At the same time, it maintains its boundary, repairs damage, and continuously readjusts its internal relationships as environmental conditions change, while the relationships it forms with its surroundings are also repeatedly reconstructed.

Stones, stars, and living organisms are plainly different, but their difference need not first be expressed by saying that one is alive while another is dead. At a more fundamental level, they differ in the way their internal and external relationships are organized, in the complexity of those relationships, and in the extent to which those relationships are continuously reconfigured. DISTANCE approaches this difference through the concept of **Dynamicity**.

Dynamicity does not simply mean moving more. It is not a matter of traveling rapidly or displaying vigorous visible motion. It refers to the structural dynamism through which numerous Momentum paths within an Island are formed in diverse ways, affect one another, connect internal and external relationships, and undergo continuing reorganization. A stone possesses Dynamicity, although the relational network through which it is expressed is relatively simple and its transformation appears extremely slow at the human observational scale. A star displays a much higher level of Dynamicity: its enormous internal configuration is continually rearranged, vast amounts of energy are released, and new relationships are formed with numerous surrounding structures. Living organisms display yet another dimension of Dynamicity. They take in matter and energy from outside, integrate them into their own configurations, continuously regulate countless internal relationships, and generate new functions when conditions require them. As the environment changes, their internal organization changes with it, and those internal changes in turn reshape their relationships with the outside.

The distinctive character of living organisms, therefore, does not lie simply in the quantity of their activity. It lies in the way numerous relationships remain tightly interconnected within a boundary while being continuously reorganized without immediately losing that boundary itself. From the perspective of DISTANCE, life is consequently not a special domain in which an entirely new principle suddenly begins to operate. It is a structural state in which the same processes present throughout the Universe appear with exceptionally high levels of organization and continuity.

Death can then be viewed differently as well. Death is not the moment at which Momentum disappears. Momentum associated with the relationships within and around an organism remains present after death. What ceases to be maintained is the organization that integrated numerous Momentum paths into a single living Island. Once that organization can no longer persist, the relationships that constituted the organism begin to be reorganized along different directions. The organism does not move from activity into an absolute state of stillness; it undergoes a transition from one organized Momentum configuration into other configurations.

The beginning of life can likewise be approached without imagining a moment at which activity suddenly emerged from entirely inactive matter. The materials composing a living organism already existed as Islands before they became parts of what we call life. They maintained internal bindings, participated in relationships with surrounding structures, and were already involved in the continuing processes of Equalization throughout the Universe. What changed was not the existence or absence of activity, but the level at which relationships became organized. At some point, numerous internal and external relationships began to form a far more highly organized configuration, one capable of maintaining an effective boundary while continuing to exchange matter and energy with its surroundings. Such configurations gradually developed the capacity to preserve their boundaries, repair damage, reorganize internal relationships, and transmit their modes of organization into other configurations. We call the region in which these characteristics become strongly concentrated **life**.

This does not mean that life is unreal merely because its category depends partly on observation. A category created by an observer is not necessarily arbitrary. Consider mountains and plains. We distinguish them without difficulty, yet nowhere in the Universe is there an absolute line declaring that a mountain ends at one point and a plain begins at the next. The terrain is continuous, and we assign different names to regions in which particular characteristics become concentrated. The distinction between a child and an adult is similar. No one becomes an adult from one moment to the next in the biological process of growth, even though societies draw practical boundaries within that continuous transformation. Day and night are not divided by an absolute line either; dawn and twilight prevent such a boundary from existing in the simple form our words suggest, yet the distinction between day and night remains meaningful.

Life need not be different in this respect. It refers to a structural characteristic that genuinely appears in the Universe, but not because the Universe has divided everything absolutely into the living and the dead. Rather, configurations with particularly high levels of structural dynamism and relational organization form a recognizable region within a broader continuum. What we call life is a densely populated region within the continuous structural spectrum of the Universe.

From the perspective of DISTANCE, the Universe does not suddenly begin applying a new principle when life appears. The emergence of life does not cause the Universe to begin following a different set of rules. Distance remains. Momentum remains. The processes of binding, separation, and Equalization do not disappear. What changes is the complexity with which they are organized, the diversity of relationships through which they become effective, and the continuity with which those relationships reconstruct one another. Life is therefore not an exceptional phenomenon inserted into an otherwise nonliving world. It is one of the regions in which the structural Dynamicity possible within the same Universe reaches an exceptionally high level.

Death, conversely, need not be placed at the metaphysical opposite of life as an absolute state. It is a process in which the organization maintained by an Island can no longer persist and is consequently reconfigured within other relationships. We experience life and death as an absolute boundary largely because we observe the world at a scale where the behavior of an integrated organism is especially easy to recognize. We see movement and sense life. We see response and judge that something is alive. We see growth and recognize the continuation of life, and when those visible characteristics disappear, we call the transition death.

The Universe, however, is not divided according to the categories through which humans most readily perceive it. Some Islands disappear almost instantaneously, while others maintain recognizable configurations for billions of years. Some reorganize themselves through explosive change, while others transform so slowly that humans can barely perceive the process. Some maintain relatively simple internal relationships, while others sustain countless Momentum paths simultaneously within a single boundary. Life is not an endpoint among these many configurations. It occupies a region of especially high Dynamicity within a vast continuum.

The first question for a philosophy of life, then, may not be exactly where life begins. A more fundamental question comes before it: why have we come to believe that the Universe itself places an absolute boundary precisely where we distinguish life from death? Once that question is opened, life no longer needs to appear as an exception inserted into the Universe. It can instead be understood within the continuous processes through which the Universe is persistently reorganized.

2 All Existence Is a Momentum Island

If the boundary between life and nonlife is not absolute, another question naturally follows. What, then, do a stone and a tree, an animal and a star, have in common? At first glance, comparing them within a single category may seem unreasonable. A stone appears almost motionless, a tree grows, an animal moves and responds, while a star releases almost unimaginable amounts of energy and influences structures across enormous distances. Their scales, structures, durations, and visible activities differ radically. DISTANCE, however, does not begin by asking whether they perform the same functions. It begins with a more fundamental question: on what basis do we recognize any of them as a single entity?

Matter gathered in one place does not automatically constitute one existence. We recognize something as a distinct entity when the many relationships composing it are bound together strongly enough, relative to their surroundings, to maintain a distinguishable configuration for some duration. What we ordinarily call a "thing" need not therefore be understood as a self-contained entity to which relationships are subsequently added. It can instead be understood as a configuration in which numerous relationships have become organized with sufficient persistence to be recognized as one structure. DISTANCE calls such a configuration an **Island**.

An Island is not something completely separated from its surroundings, nor does its boundary have to be perfectly closed. What matters is that its internal relationships are organized tightly enough to sustain, for some duration, a configuration distinguishable from neighboring configurations. A stone is an Island because its minerals and crystalline structures remain sufficiently bound to preserve a relatively stable form, even as weathering and erosion slowly alter it. A tree is also an Island: its cells and tissues, the movement of water and nutrients, and the relations among leaves, stems, branches, and roots are integrated into a distinguishable configuration that persists while matter and energy are continuously exchanged with the outside. An animal likewise consists of countless cells and organs, circulatory and nervous systems, sensory and motor structures, all connected closely enough that we recognize the organized whole as one organism. A star differs enormously in scale and mechanism, yet it too persists as a distinguishable configuration through the continuing interaction of gravitational, pressure, radiative, nuclear, and other physical processes.

These are clearly not identical kinds of existence. What they share is not a common function but a common structural condition: each persists through binding, each remains distinguishable through organized relationships, each changes as those relationships change, and none stands outside the countless coexisting Momentum of the Universe. From this perspective, what we call an entity is not an isolated object that happens to enter into relationships, but the observable persistence of a relational configuration. An Island is therefore not simply another word for "object." It identifies the structural result through which multiple relationships become sufficiently organized to maintain a distinguishishable configuration.

The importance of this distinction becomes clearer once Momentum is considered. An

Island is never merely a static arrangement of connected parts. The relationships through which it persists coexist with numerous Momentum that constrain, reinforce, cancel, and rearrange one another. Some of these relationships contribute to the relative persistence of the Island, while others alter its internal configuration or connect it to broader changes around it. The Island we observe at any moment is therefore not evidence that Momentum has ceased. Its persistence is itself compatible with continuing Momentum whose changing balance allows a recognizable configuration to remain.

A stone lying in a field illustrates this easily because its visible stability can be mistaken for inactivity. Heat is continually exchanged with the air and ground, moisture enters small fractures, temperature changes produce expansion and contraction, pressure gradually alters internal structures, and water, oxygen, and microorganisms participate in chemical processes that transform its minerals. On a human observational scale these changes may be so slow that the stone appears almost unchanged, but its persistence does not mean that Momentum is absent. It means that numerous internal and external Momentum coexist in a configuration whose overall change remains relatively limited for the scale and duration through which we observe it.

A star presents an almost opposite appearance. Its internal energy differences are enormous, matter is bound at extreme densities, nuclear reactions continue within its interior, and energy is transferred outward through radiation and particle processes while gravity participates in maintaining the large-scale configuration and in shaping relationships with surrounding Islands. Yet a star is no more an entirely independent entity than a stone. What we recognize as one star is a vast Momentum Island whose countless internal and external relationships remain organized in a sufficiently persistent configuration for us to distinguish it from what surrounds it.

A living organism belongs to the same relational Universe, although the organization within its effective boundary is far more intricate. A cell maintains a boundary while exchanging matter and signals with its surroundings. Tissues and organs continuously coordinate different flows of matter, energy, and information. Metabolism decomposes, combines, stores, and releases structures while redistributing relationships across multiple scales. The nervous system connects distant internal regions, sensory systems allow external Difference to participate in changes within the organism, and muscular activity can transform those internally organized changes into observable action upon the surroundings. None of these processes requires Momentum to be uniquely created by life. What distinguishes the organism is the extraordinary number and diversity of Momentum paths integrated within one persistent effective boundary.

This distinction is essential. If the mere presence of Momentum were taken as the criterion of life, every changing configuration would have to be called living; if nonlife were defined by the absence of Momentum, there would be no genuinely nonliving region of the Universe in that sense, because Momentum is always present. The difference between life and nonlife must therefore lie elsewhere—not in whether Momentum exists, but in how coexisting Momentum become effective within a particular relational configuration, how densely their paths are interconnected, and how persistently those relationships can be reorganized while the Island remains distinguishable.

A stone may maintain a stable configuration for a very long time, yet the number and variety of pathways through which its internal organization can be reconfigured are relatively limited. Fire can transform its surroundings rapidly, but it does not ordinarily maintain a persistent internal boundary with the organizational continuity characteristic of an organism. A star may involve vastly greater energies and physical Momentum than any living organism, yet the organization of its internal relationships is fundamentally different from that of life. A living organism, by contrast, continuously distinguishes internal from external conditions, admits some structures while excluding others, repairs damaged relationships, and modifies its own internal organization as environmental conditions change. The important question is therefore not how large its Momentum is, but how numerous Momentum are interconnected, constrained, and reorganized within a persistent configuration.

Seen this way, a living organism begins to appear not simply as a strong or active structure but as a highly organized **Momentum Island**. This also prevents the concept of Island from collapsing back into an ordinary synonym for object. What appears to us as an object cannot be adequately described without the relationships through which its distinguishable configuration is sustained. A stone cannot be separated conceptually from the mineral and crystalline structures, bindings, exchanges, and external conditions through which its present configuration persists. An organism cannot be separated from the cellular processes, chemical reactions, electrical signals, material exchanges, and environmental interactions through which its organization is continuously maintained. A star likewise cannot be detached from the physical relationships that constitute and sustain its observable structure.

In this sense, existence does not merely possess relationships as secondary attributes. What becomes observable to us as a distinct existence is expressed through a configuration of relationships sufficiently persistent to establish an effective boundary. This claim does not require the stronger assertion that relationships exist in some final ontological sense before anything else; within the scope of DISTANCE v1.1, it is enough to observe that what we identify as an Island cannot be separated from the relational configuration through which it is distinguished and maintained.

The boundary of an Island is therefore not necessarily absolute. A stone lying in a field may be recognized as one Island, yet at another observational scale it can be understood as an aggregation of different minerals, each of which can in turn be resolved into smaller crystalline configurations and still smaller structures. A forest offers the same ambiguity in the opposite direction. It can be viewed as a collection of individual trees, but if trees, roots, fungi, soil, water, animals, and atmosphere are considered as an interconnected network, the forest as a whole may also be treated as a larger relational configuration. A human being is one Island at one scale, while also comprising organs, tissues, cells, and numerous smaller structures that can themselves be treated as Islands.

Which configuration we recognize as one Island can therefore depend on observational scale and purpose, but this does not make the Island a fiction produced solely by the observer. Binding has real effects. Exchanges of matter and energy occur. Boundaries alter which interactions become effective and how strongly. What changes with observation is not whether relational organization exists, but the scale at which we choose to treat one sufficiently bound configuration as a distinguishable whole. For this reason, an Island is better understood through an **effective boundary** than through an absolute one.

An effective boundary marks the range within which relationships are organized strongly and persistently enough, relative to surrounding relationships, for a distinguishable Momentum configuration to appear. Such a boundary may be strong or weak, enduring or temporary, highly restrictive or relatively open to exchange. It may expand as a configuration grows, divide into multiple configurations, merge with others, or disappear when the organization that sustained it can no longer persist. The boundary of a stone differs from that of a cell, the boundary of a cell from that of an animal, and the boundary of an animal from those of a population or ecosystem. Yet the common feature remains that some relationships are sufficiently organized relative to their surroundings to sustain a recognizable configuration.

No Island, however, is completely isolated. Internal binding is necessary for a distinguishable configuration to persist, but persistence does not imply disconnection from the broader relational network. A stone remains exposed to heat, pressure, water, and chemical reactions. A cell exchanges molecules and signals. An animal consumes other organisms, alters its environment, and is affected by the changes it helps produce. A planet participates in radiative and gravitational relationships, while a star remains embedded within larger structures through exchanges and interactions extending far beyond its visible boundary. Every Island maintains some degree of internal organization while remaining connected to relationships beyond itself, and because those external relationships continue to change, its internal configuration can never be regarded as permanently fixed.

This dual character becomes exceptionally elaborate in life, but it does not begin with life. A living boundary is not a wholly new principle inserted into the Universe. It is an extraordinarily intricate form of a more general structural condition: the maintenance of an effective boundary amid continuing interaction. In living systems, binding becomes multilayered, boundaries become increasingly selective, and internal relational networks become vastly more complex. External Differences can participate in many interconnected internal changes, while the resulting configuration can in turn alter its relationships with the environment. The Island consequently does more than merely persist; its relational organization can be repeatedly reconstructed as surrounding conditions change.

Even this complexity does not separate life from the rest of the Universe. The matter composing living systems comes from other Islands, and the Momentum effective within them remains part of the same broader network of Difference, Distance, Momentum, and Equalization already present throughout the Universe. A living boundary is not a wall that removes an organism from that network. It is an effective boundary through which exchanges with the outside are constrained, differentiated, and organized. When that organization can no longer be maintained, the smaller Islands that had participated in the organism become incorporated into other relational configurations. Life does not enter from outside the Universe, nor does death carry anything beyond it.

What remains are Islands that bind, persist, exchange, separate, and enter new configurations. A stone, a star, and a living organism do not belong to three disconnected worlds; they are differently organized Momentum Islands within the same relational Universe. Their decisive difference is therefore not whether Momentum exists, but how densely, diversely, and persistently coexisting Momentum are organized and reorganized within their effective boundaries. It is along this dimension of structural dynamism that the differences among Islands begin to form a spectrum rather than an absolute divide.

3 Structural Dynamicity Created through Momentum Resolution

If every existence can be understood as a Momentum Island, the difference among a stone and a flame, a cell and a tree, or a human being cannot be explained by saying that some possess Momentum while others do not. Every Island participates in Momentum, maintains internal relationships, remains involved with structures beyond its effective boundary, and undergoes formation, persistence, transformation, and eventually the loss of the organization by which it had been recognized as a distinguishable structure. The difference lies not in whether Momentum exists, but in how it is organized and how its resolution proceeds. DISTANCE calls this difference **Dynamicity**.

Dynamicity does not simply mean that something moves a great deal. An object may travel at very high speed while its motion is governed by only a few relatively simple Momentum paths. Conversely, a dormant organism that appears almost motionless may maintain numerous internal relationships capable of becoming effective in many different directions when surrounding conditions change. Speed, therefore, cannot serve as a measure of Dynamicity. Dynamicity concerns not the rapidity of motion but the organization of the relational network through which different Momentum paths can become effective.

Nor does Dynamicity refer to the magnitude of physical momentum or energy. A star can involve incomparably greater physical momentum and energy than a living organism, a vast storm can transform an entire forest, and an explosion can produce more visible change in a fraction of a second than a single cell does over a much longer interval. Yet none of these comparisons by itself tells us which structure has greater Dynamicity. The relevant question is how many different Momentum paths are organized within and across an Island, how closely those paths are connected, and how changes in one path alter the conditions under which others become effective.

Dynamicity is not exclusive to life. Stars maintain countless internal relationships; atmospheric systems sustain complex circulations; ocean currents interact continuously and generate new patterns of change; chemical reaction networks contain multiple pathways that affect one another; planetary systems preserve many interacting relationships over immense durations. Each displays Dynamicity according to the way its Momentum paths are organized. Dynamicity is therefore a structural characteristic found across Islands, although the ways in which Momentum becomes effective, interacts, changes direction, and is resolved differ greatly from one Island to another.

From this perspective, Dynamicity can be understood as **the degree of structural dynamism that appears as an Island, within and across its effective boundary, organizes multiple Momentum paths, connects them with one another, redirects them, and resolves them through changing relationships**. The word *degree* is important. Dynamicity is not a property that some entities possess and others lack. Every Island exhibits it at some level. The question is not whether Dynamicity exists, but how diverse the Momentum paths organized within a structure are and how they are related to one another.

The number of paths alone, however, is not sufficient. Consider a falling stone. The stone is unmistakably involved in Momentum relationships: its position changes in relation to Earth, friction with the air generates heat, and when it strikes the ground both the stone and the surface can be reorganized. Within the stone, atoms and molecules continue to interact, and the impact may alter its internal structure. None of this is static. Yet the range of Momentum relations that can be organized through the stone remains comparatively constrained by its existing structure. The stone does not sense the approaching ground in advance, reorganize itself in anticipation of impact, select a new path in order to repair damage afterward, or alter its organization so that a later collision will be handled differently because of the present one. This does not mean that the stone is inactive or lacks Momentum; it means that the range of Momentum paths that can become organized through its present structure is relatively limited.

A cell presents a different configuration. It receives many kinds of chemical Difference from its surroundings. Receptors at its membrane interact selectively with external structures, activating different internal pathways. Some molecules enter the cell, some are decomposed, some are incorporated into larger structures, some are stored, and others become signals that alter still other Momentum paths. When damage occurs, repair processes can be activated; when environmental conditions change, membrane selectivity and internal regulation can also change. Genetic and regulatory systems continually affect which pathways remain active and which are suppressed. What matters here is not merely that many processes occur at once, but that they are interconnected. A change in one relationship can modify the conditions under which numerous other relationships become effective.

This connectivity distinguishes Dynamicity from mere multiplicity. Hundreds of processes occurring simultaneously but independently may produce a complicated scene without producing correspondingly high Dynamicity. If, however, a small change in one relation modifies the direction, efficiency, or accessibility of many other Momentum paths, the entire structure can be reorganized through that change. Dynamicity therefore concerns not simply the number of Momentum paths but the density of their interconnection and the extent to which one change can systematically reorganize many others.

For the same reason, Dynamicity is not identical with **complexity**. A structure may contain a very large number of components while those components remain only weakly related to one another. Such a structure may be complex in composition without exhibiting particularly high Dynamicity. Conversely, a structure containing fewer components may display much greater Dynamicity if changes among them propagate through a tightly connected relational network and repeatedly reorganize the whole. Dynamicity concerns less how much is present than how extensively relationships organize and reorganize one another. In this sense, high Dynamicity requires both **multiplicity** and **integration**.

Even multiplicity and integration are not enough by themselves. A complex Island must also be able to reorganize the paths through which Momentum is resolved as its relationships change. Momentum is not resolved only along one permanently fixed route. Within a complex Island, a Momentum path may be delayed, redirected, divided into several branches, locally amplified, or connected through another structure into a different direction. Living organisms provide particularly clear examples. A plant transforms incoming radiant energy into chemical binding; an animal can transform an external chemical Difference into electrical signaling within its nervous system; the nervous system can connect those signals to muscular movement; and that movement can then create external relationships that did not previously exist. No new Momentum suddenly comes into existence in this sequence, nor does Momentum disappear. What changes is the organization of the paths through which it becomes effective and is resolved.

Higher Dynamicity therefore means that such pathways can be reorganized with greater diversity and refinement. The more broadly an Island can reconstruct its Momentum paths as surrounding conditions change, the greater the Dynamicity it can display. This also distinguishes Dynamicity from simple **reactivity**. Every Island responds in some manner to external change. Metal expands when heated, a crystal fractures under sufficient pressure, and water changes phase according to temperature and pressure. Such responses can be substantial, but the response itself does not necessarily reorganize how the structure will respond to later Differences.

In an Island with higher Dynamicity, by contrast, one response can modify the structure of subsequent responses. An incoming Difference changes several internal relationships, and those changes alter how the next Difference will become effective. An interaction therefore need not end with its immediate consequence; it may leave a structural result that continues to affect many later interactions. What is commonly called memory need not begin with consciousness. A structure can retain the effects of previous interactions without being conscious of them. A cell's later responsiveness can change because of earlier chemical interactions; an immune system can respond differently after prior exposure to an external structure; tissues can become more strongly bound, or weakened, through repeated stimulation. Across living systems, previous Momentum relationships frequently alter the paths through which later Momentum becomes effective.

This principle is not confined absolutely to life. Repeated pressure can gradually alter the internal arrangement of a material, and repeated heating can change its subsequent responses. What matters is not the scale of the change but the **structural continuity** through which earlier relationships continue to modify later ones. DISTANCE does not treat time itself as the cause of this continuity. The past affects the present not because time acts upon a structure, but because earlier interactions have left actual changes within the structure, and those changes remain present in the relationships through which current Momentum becomes effective. What we call the influence of the past is therefore the continued effectiveness of a structure produced by earlier Momentum relationships.

Dynamicity consequently includes more than the simultaneous maintenance of many Momentum paths. It also includes the structural continuity through which earlier Momentum relationships can continue reorganizing later paths. An Island does not need to "store the past" as something separate from its present configuration; the effects of previous interactions remain effective because they have become part of the current structure. One change can therefore modify another, which in turn changes subsequent relationships, allowing Dynamicity to develop beyond immediate reaction into continuing structural reorganization.

Such reorganization, however, requires conditions under which it can persist. Momentum paths cannot remain interconnected, nor can the effects of earlier changes remain structurally effective, unless an Island maintains enough continuity to remain a distinguishable configuration. Dynamicity is therefore deeply connected with the **Effective Boundary**. If an Island cannot maintain sufficient internal binding, its relationships disperse before multiple Momentum paths can accumulate, connect, and be reorganized within a continuing structure. Yet if the Island were to exclude every external relationship completely, new Difference could no longer become effective through its internal organization. High Dynamicity develops between these extremes: the Island must remain sufficiently bound to preserve internal relationships while remaining sufficiently open for external Difference to participate in their continuing reorganization.

An Effective Boundary is therefore more than a wall separating inside from outside. It organizes how external relationships participate in the Island: which interactions are admitted, which are delayed, which are constrained, and which can be transformed into different Momentum paths. The cell membrane provides a particularly clear biological example. A cell does not admit every surrounding structure indiscriminately. Some materials pass selectively through its boundary, while others are excluded or processed through different pathways, and these selective relations become part of the organization through which the cell maintains its internal Momentum structure.

Selective boundaries, however, do not begin with life. Every Island possesses an Effective Boundary in some form, although the degree and organization of selectivity differ greatly. A stone exchanges heat with its surroundings but does not respond identically to every influence. A crystal grows according to particular structural relations rather than combining randomly with every surrounding material. A planet receives radiation and participates in gravitational relationships while maintaining an internal configuration through which those influences become effective in different ways. What distinguishes highly Dynamic Islands is not the existence of a boundary but the increasingly intricate manner in which the boundary participates in organizing relationships.

For this reason, Dynamicity must not be identified with structural stability. A highly stable crystal can preserve its form for a very long time while offering relatively limited possibilities for the formation and reorganization of new internal relationships. A living organism persists in almost the opposite manner. Cells are replaced, molecules are broken down and recombined, tissues undergo damage and repair, and countless internal relationships are continually modified. Yet we continue to recognize the organism as the same Island because its persistence does not depend on every component remaining unchanged. What persists is the organization connecting those changing components.

This reveals a distinctive form of stability. An Island with low Dynamicity may persist because relatively little changes within it, whereas an Island with high Dynamicity can persist through continuous change. New relationships are formed, existing relationships are modified, damaged structures are repaired, and new Difference becomes connected to further Momentum paths. Change is not merely something that threatens the structure;

organized change can become the very means by which the structure continues. Living organisms display this especially clearly. They persist not simply by resisting change, but by organizing it.

This also makes it unnecessary to portray life as an exception that preserves order by standing against the broader processes of the Universe. In the DISTANCE interpretation, living structures remain within Equalization rather than opposing it. Their dense internal relational networks can provide additional ways through which existing Difference becomes connected to Momentum paths and through which Momentum Resolution proceeds. The persistence of a living Island and the broader processes of Equalization are therefore not inherently opposed. A living Island is a local Momentum structure formed and maintained within those broader processes, while its internal organization can diversify the paths through which surrounding Differences become involved in further change. High Dynamicity does not stop Equalization; it can multiply the relational pathways through which resolution becomes possible.

Yet increasing structural complexity does not automatically produce increasing Dynamicity. A system may contain many components while repeatedly using only a narrow set of relationships, whereas a smaller system may generate diverse Momentum paths through a tightly interconnected network. The number of components, the amount of matter, the magnitude of energy, and the speed of visible motion are therefore insufficient as measures. Nor should Dynamicity be treated as a scale of evolutionary superiority. A bacterium, a whale, a forest, and a human society can each exhibit high Dynamicity in very different ways. A later evolutionary appearance does not by itself imply a more Dynamic structure, nor does greater physical size.

Dynamicity can also vary within what we regard as the same Island. A dormant seed and a germinating seed belong to one structural continuity, yet the number and interaction of Momentum paths presently organized within them can differ greatly. A virus outside a host may display only a restricted range of effective relations, while entry into a suitable cell allows many previously unavailable pathways to become organized through new relationships. The same organism can display markedly different Dynamicity in sleep and wakefulness, dormancy and activity, growth and decline. This does not mean that it becomes a fundamentally different kind of existence at every moment. Dynamicity is not a fixed intrinsic quantity but a structural characteristic that appears through relationships.

What an Island can actually organize is therefore not determined by internal structure alone. It also depends on what other Islands surround it, what Differences are present, and which new relationships can become effective. The same seed placed on dry rock and in moist soil remains structurally related to the same seed, yet the relationships available to it are radically different, and so are the Momentum paths that can become organized. External relationships make some possibilities within a structure effective while leaving others unrealized. Dynamicity should therefore not be understood simply as a capacity stored inside an entity. It emerges through the relation between internal organization and external conditions.

An Island may contain many Momentum paths that are not presently effective. Without appropriate external relationships, they may remain unavailable within the observable organization of the structure; when new relationships form, however, pathways that had remained only structurally possible can become effective and participate in reorganizing the Island as a whole. Dynamicity consequently includes not only the Momentum structure presently expressed but also the breadth with which different Momentum paths can become organized as relationships change.

Taken together, these features allow a more precise understanding of the concept. **Dynamicity is the degree of structural dynamism through which an Island, while maintaining its Effective Boundary, continuously organizes diverse paths of Momentum Resolution, connects them with one another, and reorganizes them as its relationships change.** A structure in which virtually no change occurs would offer little opportunity for new Momentum paths to become organized. At the opposite extreme, if all binding disappeared and every relation dispersed, no Island would remain within which such paths could be connected. Dynamicity emerges between these extremes, where Difference remains, relationships connect those Differences, and sufficient structural continuity allows one change to participate in further changes.

Every Island occupies some region within this continuum. Stones and flames, crystals and viruses, seeds and cells, animals and humans do not need to be divided first into fundamentally disconnected kinds in order for their differences to be recognized. They are Momentum Islands displaying persistence, exchange, integration, reorganization, and structural continuity in different combinations and to different degrees. What we call life occupies a region in this broader continuum where Dynamicity becomes especially dense and highly integrated.

Dynamicity does not begin with life. It is a structural characteristic that can be observed, in different forms and degrees, across Islands. The more important question is therefore not which entity first acquires Dynamicity, but why the pathways of Momentum Resolution become extraordinarily diverse, interconnected, and continuously reorganized in some Islands, and how such configurations come to occupy the region that we recognize as life.

4 The Continuous Spectrum of Islands

Once Dynamicity is understood as the degree to which diverse paths of Momentum Resolution are organized and reorganized within an Island, the boundary that once seemed to divide life from nonlife no longer appears absolute. The Universe does not consist of two separate domains, with static matter devoid of Momentum on one side and living beings governed by an entirely different principle on the other. It consists of countless **Islands**. Every Island is formed through internal binding, maintains an effective boundary for some duration, exists in relation to surrounding Islands, and displays Dynamicity to a different degree. What becomes most important here is the idea of **continuity**.

Continuity does not mean that all Islands are alike. A stone differs from a flame, a virus from a bacterium, and a tree from a human being. Their internal organizations differ, as do the ways in which they form relationships with their surroundings, the durations for which they persist, and the patterns through which they change. Yet explaining these differences does not require us to divide existence itself into two fundamentally separate ontological domains. Between one form of organization and another lie innumerable intermediate configurations rather than an absolute discontinuity. From structures that appear almost unchanged to those that maintain extraordinarily complex forms of self-organization, Islands occupy different regions of a continuous spectrum. The same continuity persists even around the boundary that we ordinarily draw between life and nonlife.

A crystal incorporates surrounding material and grows into an ordered structure. A flame consumes external material and continues to extend its form as long as appropriate conditions remain. A vortex preserves a recognizable configuration for some duration even though the matter composing it is continually replaced. A virus may show almost no independent activity outside a host, yet once inside a suitable cell it can activate complex Momentum paths. A dormant seed may appear almost unchanged for a long period while preserving an intricate organization capable of later reactivation. A cell continuously exchanges matter and energy with its surroundings while maintaining its structure, yet the collapse of only a few critical Momentum paths may rapidly destroy its integration.

The point of these examples is not that all such structures should be called living. Rather, the characteristics by which we identify life do not all begin simultaneously at a single absolute boundary. One structure may grow without engaging in metabolism in the biological sense. Another may replicate without continuously maintaining itself. A structure may form a boundary without possessing a genetic system; complex chemical exchange may occur without a nervous system; responsiveness to external change does not necessarily imply consciousness; and a structure may persist for a long period while exhibiting almost no visible motion. The properties associated with life are distributed differently among different Islands rather than appearing together at one sharply defined point. It is therefore more natural to understand the region we call life not as the point where one decisive property suddenly appears, but as a region in which multiple forms of Dynamicity begin to become tightly organized beyond a certain level. The spectrum of Islands is consequently not a sequence of separate boxes, but a continuous structure across which the density, diversity, and structural integration of Dynamicity change gradually.

The spectrum of Dynamicity should therefore not be imagined as a single straight line running from a lower stage to a higher one. Although every Island exists somewhere within a continuous spectrum, the structure determining its position is far from simple, because Dynamicity is not determined by one factor but emerges from multiple, tightly interconnected structural characteristics. One Island may maintain exceptionally strong internal binding while having only a limited capacity to adapt to external change. Another may change rapidly but fail to preserve continuity as a single structure for long. Some Islands may be highly sensitive to external Difference, while others may coordinate numerous internal relationships with great precision. Some organize exchange with the outside through highly selective boundaries; others can restore damaged bindings, allow previous relationships to remain reflected in later configurations, or extend their mode of organization into new Islands.

These characteristics do not exist independently. Within an Island they influence one another and form many different combinations. Strong internal binding does not necessarily imply high adaptability, just as high adaptability does not necessarily imply complex structural memory. A large number of internal relationships does not always produce a strong capacity for self-maintenance, while a comparatively simple structure may display highly stable Dynamicity under particular environmental conditions. Dynamicity should therefore not be treated as a simple scale expressible by a single number. Two Islands that appear to display a similar degree of Dynamicity may in fact sustain their structures in entirely different ways. One may obtain stability primarily through strong internal binding, another through continuous exchange and reorganization, and still another through heightened sensitivity to external change. What appears to occupy the same region of the spectrum may thus arise from very different structural combinations.

For this reason, the Dynamicity spectrum is not intended to assign every Island a fixed numerical value, nor is its purpose to rank all forms of existence along a single metric. It offers instead a conceptual continuum through which we can abandon the assumption of an absolute discontinuity between life and nonlife and recognize that every Island organizes structural dynamism in a different way. Even the apparent position of an Island within that continuum can change according to the scale at which it is observed.

Watch a stone in a field for a few minutes and almost nothing appears to happen. At the atomic and molecular scales, however, interactions continue, while across geological scales fractures develop, chemical reactions proceed, weathering and erosion transform the stone, and its material eventually becomes incorporated into other structures. The Dynamicity that becomes visible therefore differs according to the scale at which its Momentum relationships are observed. Living organisms reveal the same dependence on scale. Viewed as a whole, an animal may appear far less active while asleep, even though metabolism, chemical reactions, and electrical signaling continue at the levels of organs and cells. Conversely, patterns invisible when attention is confined to one individual may become apparent at the scale of a population, where reproduction, competition, migration, and genetic change form a larger Momentum structure.

What counts as one Island can therefore vary with observational scale. A cell can be treated as an Island, while the organism composed of those cells can also be treated as a larger Island. A population or an ecosystem may likewise be understood as an Island when its internal relationships are sufficiently organized. This does not make Dynamicity arbitrary. Real relationships exist at every scale; what changes is the level of the relational network selected for observation as one Island. Dynamicity is thus not merely a quantitative comparison but a structural characteristic shaped jointly by relational organization and observational scale, and this is one reason all Islands must be understood within a continuous spectrum.

An Island's position within that spectrum can also vary with the conditions and duration under which it is observed. A flame exhibits violent change over a short interval, whereas a mountain range may appear almost unchanged over the span of a human life.

Across millions of years, however, the same mountain range continually rises and erodes while forming new relationships with surrounding structures. A seed that has remained dormant for a long period may begin complex biological activity within a relatively short interval once appropriate conditions arise. Even in an organism approaching death, internal relationships do not vanish at one moment; different structures lose their integration at different rates. A single snapshot is therefore insufficient to characterize an Island's Dynamicity. A structure that appears to display very low Dynamicity at one moment may, under suitable conditions, organize numerous Momentum paths that had not previously become observable.

Dormant biological structures make this particularly clear. A dormant seed is not simply a dead object waiting to become alive again. It preserves structural possibilities that can later become effective. Water, temperature, the chemical conditions of the soil, and light can establish external relationships through which previously insufficiently connected structures become active and new Momentum paths become effective. The resulting change may appear sudden, yet nothing has emerged from an empty structure. Possibilities maintained within the existing organization of the seed have become effective through newly established relationships.

Viruses pose a similar problem. Outside a host, a virus exhibits little independent activity, but once inside a suitable cell it begins to connect with the many Momentum paths already organized by the host. Replication, transformation, and propagation that were previously unavailable can then occur. Asking whether a virus is alive or not may therefore conceal a more fundamental issue. A virus does not become an entirely different kind of existence at the moment it enters a host. What changes is the structure of its relationships: by forming new relations with the host, Momentum paths that were previously ineffective become organized and effective.

This principle is not confined to viruses. The Dynamicity of any Island is not determined by internal structure alone. Even an apparently identical structure can organize very different Momentum paths depending on the environment in which it is situated. The same bacterium in a nutrient-rich environment and under severe deprivation can display very different Dynamicity. The same plant exposed to abundant sunlight or deprived of light can organize Momentum in very different ways. An animal embedded in social relationships and the same animal in complete isolation can display markedly different structural dynamism through behavior, information exchange, and interaction with its surroundings. Dynamicity is therefore not a property fixed and stored inside an Island.

It is relational, arising through the interaction of internal organization and external relationships.

An Island may contain numerous Momentum paths that are not presently effective. Until appropriate relationships are formed, those paths may remain unexpressed within the observable configuration. Once new relationships arise, however, pathways that previously existed only as structural possibilities can become effective and reorganize the Island as a whole. An Island therefore cannot be understood solely through what is visibly active at the present moment. Alongside already organized Momentum relationships, it contains structural possibilities that may become effective through future relationships. Dynamicity encompasses not only present activity but also the diversity of Momentum paths that can become organized as relationships change. The position an Island occupies within the spectrum should accordingly be understood not as a fixed value but as a structural characteristic that changes with relationships and conditions.

Because Dynamicity varies relationally in this way, it must not be interpreted as an evolutionary stage or a hierarchy of value. Higher Dynamicity does not mean a superior form of existence, just as lower Dynamicity does not mean a less complete one. The Universe does not arrange all Islands toward a single destination. Each forms differently, persists differently, and organizes Momentum differently. A star involves vastly more matter and energy than a cell, and a storm can reorganize structures across an enormous region, while a bacterium, despite its minute size, can respond sensitively to environmental change and continuously readjust its internal relationships. These are different forms of Dynamicity; none establishes that one Island is intrinsically a higher form of existence than another.

Greater complexity can increase the range of possibilities, but it also increases dependency. The addition of a new Momentum path can provide a new possibility for adaptation while simultaneously introducing another path through which failure can occur. The more tightly connected a structure becomes, the more likely it is that a small change can influence the configuration as a whole. High Dynamicity provides more possibilities, but more possibilities also bring additional constraints and risks. Dynamicity is therefore not a ladder indicating a direction of progress. It is a **field of structural possibilities**. Some Islands persist for a long time because they change very little, whereas others preserve their organization precisely because they change continuously. Some maintain their boundaries through strong binding, while others achieve a similar persistence through selective exchange.

Some Islands resolve Momentum primarily through comparatively simple physical interactions. Others tightly connect chemical, electrical, mechanical, and biological relationships within one structure and organize Momentum in far more complex ways. These differences are different structural forms appearing across the Dynamicity spectrum. The region we generally call life is one in which many such relationships are connected at particularly high density.

Yet even within the region called life, not every Island displays the same Dynamicity. Bacteria and dormant spores, enormous trees and insects, mammals and humans all organize Momentum differently. Their internal relational networks differ, their effective boundaries differ in character, and they form relationships with external Islands in different ways. Their capacities to maintain and reorganize structure are likewise not identical. We nevertheless call all of these structures living because, despite their differences, they occupy a recognizable region in which multiple forms of Dynamicity are tightly integrated beyond a certain level. That commonality does not erase continuity or variation. The category of life gathers many different configurations under one name; it does not make them a uniform state.

Death is likewise not a fixed state absolutely disconnected from life. When an organism dies, all Momentum paths do not disappear simultaneously, nor do all of its structures change in the same manner. Some relationships collapse first, others remain for a time, and still others become incorporated into new structures. Life and death should therefore not be imagined as opposite endpoints of a single line. Life is not the maximum value of Dynamicity, and death is not its minimum. A powerful storm or a star may display intense physical Dynamicity without being alive, while a dormant organism may appear almost unchanged even though it preserves a highly organized structure capable of becoming active again. What matters is not the quantity of activity but how that activity is organized.

The Dynamicity spectrum is consequently not a standard for evaluating or ranking forms of existence. It is a structural map for understanding how different Islands organize and resolve Momentum. On that map, the region we call life should be understood not as an exceptional insertion into the Universe but as a region in which particular forms of Dynamicity become densely organized.

The same continuity also applies within the changing condition of a single living organism. An organism does not remain at one fixed position on the spectrum throughout growth and dormancy, activity and recovery, aging and decline. Its Momentum structure continually changes, and the Dynamicity it displays changes with it. Even when its outward appearance changes little, the Momentum paths organized internally continue to be reconstructed. Conversely, a structure that appears almost inactive can display very different Dynamicity once appropriate conditions arise. A dormant seed is again illustrative: although it may appear to do almost nothing for a long period, relationships capable of later reorganization remain preserved within it. When new external relationships involving water, temperature, soil, and other conditions are established, many Momentum paths that were previously not effective become active. What looks like a sudden transition is the reactivation, through new relationships, of organization already maintained within the existing structure.

Death does not escape this continuity either. When an organism is no longer recognized as alive, not every structure changes in the same way at the same moment. Some relationships collapse earlier, some persist for a while, and others become parts of new Islands. Judging Dynamicity from a single instant is therefore insufficient. A structure that currently appears almost inactive may preserve the possibility for new relationships to become organized, while intense activity does not by itself imply the high Dynamicity of an integrated living Island if that activity is not organized in a way that maintains an integrated structure. Dynamicity cannot be reduced to the amount or speed of change. What matters is how tightly Momentum paths are connected and how continuously they can remain organized within a structure.

What ultimately matters on the spectrum is not the appearance of an Island at one moment, but the manner in which it maintains and transforms its Momentum structure through diverse relationships. The same structure can display different Dynamicity as conditions change and can occupy different regions of the spectrum while preserving structural continuity. The distinction between life and nonlife therefore cannot be settled merely by asking whether activity is present at a particular instant. The deeper question is why we recognize a particular region within this continuous spectrum as an independent category.

Seen as a whole, the spectrum reveals three related features. Every existence is a Momentum Island, participating in internal and external Momentum relationships while continually changing through its relations with surrounding Islands. Dynamicity varies continuously rather than beginning at an absolute break between life and nonlife, because the complexity of Momentum Resolution is organized differently in different Islands. And an Island's position on the spectrum is relational rather than being determined by internal structure alone: it varies with observational scale, the duration and conditions of observation, relationships with surrounding Islands, and the conditions under which presently unrealized Momentum paths become effective.

The question of life and nonlife is therefore not simply whether a structure is active or inactive. It is why Dynamicity becomes concentrated in particular ways within some Islands, and by what criteria we recognize such configurations as belonging to one category. The Universe does not produce a separate substance called life, nor does it introduce a new law for living structures. It contains countless Islands bound in different ways, forming different relationships and organizing Momentum through different structures. What we call life is a region among them in which Dynamicity displays a particularly high density and degree of integration. That region is real, but its boundary is not an absolute discontinuity; it changes gradually within a continuous spectrum.

Recognizing the reality of life is therefore different from treating life as an absolute ontological category. DISTANCE does not deny life. It recognizes the distinctive structural characteristics of living Islands while emphasizing that those characteristics do not arise from a principle detached from the rest of the Universe. Life is not an exception separated from the Universe, but a region in which relational principles common to the Universe become organized in a distinctive way. The question that remains is no longer simply **"What is alive?"** It is instead **"Why do we recognize a particular region of the continuous spectrum of Dynamicity as a distinct category?"**

5 Life as a Region Defined by the Observer

If every Island exists somewhere on a continuous spectrum of Dynamicity, life can no longer be understood as a special substance granted only to certain kinds of matter, nor as an absolute boundary dividing the Universe into two fundamentally separate domains. In DISTANCE, **life is a region**. It is a region within the continuous spectrum formed by countless Islands where Momentum paths exhibit particularly high density and integration. We repeatedly observe structures concentrated in this region and give them a common name: life.

Living organisms generally maintain an Effective Boundary while continuously exchanging matter and energy with their surroundings. They reorganize incoming structures within themselves, preserve internal relationships while continually changing, repair damaged structures, and reorganize internal and external Momentum paths as environmental conditions change. In some cases, they can also extend their mode of organization into new Islands. These characteristics actually exist; they are not illusions arbitrarily invented by human observers. Yet the boundary by which we gather these characteristics into a single category is established by the observer.

This distinction is crucial. To say that life is defined by an observer does not mean that life is merely subjective. Calling a stone alive does not make it a living organism. The point is almost the reverse. The Universe presents us with countless structures whose Dynamicity varies continuously, and the observer identifies a region in which certain structural characteristics repeatedly appear together and gives that region a categorical name. **Life is real, but its category is not absolute.**

A simple analogy helps clarify the distinction. The boundary between land and sea has real effects. The two regions contain different materials and ecosystems and undergo change in different ways, yet their boundary does not always exist as a single fixed line. Tides continually move the shoreline, waves blur it, and tidal flats, deltas, and wetlands form intermediate regions through which land and sea merge gradually. The fact that the boundary moves does not mean that land and sea do not exist. It means that their boundary is a structure whose location depends on relationships and scale.

Life can be understood in a similar way. We observe a region in which Dynamicity becomes particularly dense and tightly organized and call that region life, but as we approach its margins, the boundary becomes increasingly difficult to draw. Viruses, dormant spores, seeds, synthetic life forms, cellular organelles, and self-replicating molecules continue to complicate attempts to define life through a single criterion. They organize Momentum in different ways and occupy different positions within the Dynamicity spectrum. Some structures preserve complex information but cannot independently organize external Momentum; some maintain boundaries but cannot reproduce themselves; some can be replicated but do not sustain independent metabolism; still others display almost no activity until they form appropriate relationships with a particular environment.

The ambiguity at the boundary of life is therefore not merely a consequence of insufficient observation or incomplete biological knowledge. More fundamentally, the Universe itself does not arrange all existence into two completely separated populations. The category we call life corresponds to a structural region that repeatedly appears within a continuous spectrum, and the edges of that region need not be sharply divided.

The same problem becomes apparent when we ask when life begins. At the level of an individual organism, there may be many practically meaningful criteria for marking a beginning: fertilization, the formation of a particular cellular organization, the emergence of integrated physiological functions, independent viability, birth, or some other developmental threshold. Different disciplines and institutions may require different boundaries because they address different questions. Biology, medicine, ethics, and law need not always select precisely the same point, and the criteria they use can each have substantial practical significance. Yet there is no requirement that all of these boundaries correspond to one identical ontological instant.

What is continuous is the developmental process itself. The observer selects a particular point within that continuity as a practical criterion according to the purpose of observation. Again, this does not make the category of life arbitrary. It means that life is a real region formed upon continuous structural change, while observers distinguish and use that region according to the purposes for which a boundary is needed. Life is real, but its reality does not arise from one absolute boundary. It appears within a continuous structure in which relationships become progressively organized. Understanding the beginning of life therefore depends less on discovering one decisive instant than on understanding how relationships gradually come to organize higher levels of Dynamicity.

Categories established by observers perform an indispensable role not only here but throughout science and other forms of inquiry. Without categories, we could neither compare nor classify different forms of existence, identify common structures, nor explain why different phenomena resemble or differ from one another. The category of life serves precisely this purpose: it allows us to understand, as one meaningful region, structures in which Dynamicity achieves a particularly high density and degree of integration.

A problem arises only when we forget that such categories are tools for observation and understanding and begin to treat them as absolute divisions already inscribed into the Universe itself. Categories are established so that recurrent structural characteristics can be identified and explained. They are concepts through which structures become intelligible; they do not necessarily correspond to boundaries that the Universe itself has drawn in advance.

Life is no exception. The structures we call living genuinely exist. They maintain Effective Boundaries, tightly organize diverse Momentum paths, preserve their configurations through selective exchange with the outside, and sustain integrated structural continuity even while undergoing continual change. These characteristics provide real structural grounds for distinguishing living Islands from many other Islands. Their distinctiveness, however, arises from differences in degree and organization, not from the presence or absence of Momentum. Nor does it arise from whether a structure participates in the Equalization of the Universe. Living organisms exist within the same relational principles as other Islands.

A living organism receives external Difference and reorganizes its internal relationships through it; changes formed within those relationships can in turn alter the organism's external relationships. Matter and energy cross its Effective Boundary selectively, and the resulting Momentum paths become connected to other internal and external Momentum paths. The organism does not stand outside Equalization while preserving itself against it. Its persistence, exchange, transformation, and eventual reorganization all occur within the same broader processes through which other Islands form, persist, and change.

The category of life therefore reflects something objectively real without requiring an absolute boundary. This is the point at which the apparent opposition between objectivity and observer-dependence must be reconsidered. If life is treated only as objective, it can easily be mistaken for an absolute ontological boundary established in advance by the Universe. If observer-dependence alone is emphasized, life risks being reduced to nothing more than a name or subjective judgment. DISTANCE accepts neither extreme.

Life is not another Universe separated from nonlife, nor is it a special substance added to matter from outside. It is a region of the Dynamicity spectrum formed when Momentum and relationships become organized in particular ways. That region is distinctive enough for us to identify it with a common name, yet its boundary connects gradually with neighboring structures rather than existing in complete discontinuity from them.

Recognizing life therefore involves two aspects at once. First, **life is objective**. Living organisms actually maintain high-density Dynamicity, preserve Effective Boundaries, organize diverse Momentum paths in tightly interconnected ways, and sustain their structures through selective relationships with what surrounds them. These structural characteristics exist regardless of what an observer thinks about them. Second, **life is observer-dependent as a category**. An observer identifies the scale and degree at which these structural characteristics constitute an integrated Island and distinguishes that region under the category of life. The structure is objective; what the observer determines is the scale, purpose, and boundary through which that structure is understood as one category.

These two aspects are not contradictory. Both are necessary if the concept of life is to be understood without turning either into an absolute metaphysical division or into an arbitrary act of naming. Life is not an illusion, but neither is it an absolute ontological entity. It is a region in a continuous Universe where real structural characteristics attain a particular density and organization, and the observer identifies that region as a meaningful category.

This perspective places life back within the wider relational structure of the Universe. Life is not an exception that escapes its general principles but a configuration in which the same relational principles become organized at an especially high level of Dynamicity. To understand life, therefore, is not to search for another principle detached from the rest of the Universe. It is to understand how the same relational principles can, in particular regions, become organized into increasingly dense, integrated, and persistent forms of Dynamicity.

The same interpretation must eventually be applied to death. If life does not refer to an independent substance but to a particular state of structural organization, death cannot simply be understood as the disappearance of that substance. It must instead be approached as a transition in which an Island can no longer maintain the structural organization through which it had been recognized as living—a transition of its **Momentum Structure** into other configurations.

6 Death as a Transition of Momentum Structure

If life is not an absolute ontological category, death cannot be understood as its absolute opposite either. We commonly think of death as the termination of all activity: breathing stops, the heart no longer circulates blood, the integrated regulation of the nervous system disappears, and the body ceases to respond to its surroundings as a single individual. From the perspective of an observer who regards the entire organism as one Island, these changes are unmistakable. A structure that once moved, received sensory input, repaired damage, retained memory, and continuously organized itself no longer behaves in the same way. It is therefore easy to conclude that life existed and then, at a particular moment, disappeared.

But what actually disappears at death? The matter composing the body does not vanish, nor does energy disappear from the Universe, and Momentum does not suddenly cease to exist. The countless smaller Islands that constituted the organism do not all stop their activity simultaneously. Even after the organism is judged dead, cells can continue functioning for different periods, chemical reactions continue, heat is transferred to the surroundings, microorganisms begin forming new relationships, and molecules and atoms become incorporated into configurations different from those they occupied before. What disappears, therefore, is not activity itself but the structural organization that integrated these many activities into one living Island.

Death, in this sense, is not an event in which an active existence simply becomes inactive. It is a structural transition in which the integrated **Momentum Structure** that maintained an organism as one living Island can no longer be sustained. A living organism does not exist through one single movement; it is an immense relational network of tightly connected Momentum paths. While the organism remains alive, those paths constrain and regulate one another in ways that preserve the integrated structure. A change in one pathway can induce changes in others, while local disturbances are repeatedly readjusted within the stability of the larger configuration.

Once that integration begins to collapse, pathways that had previously been connected within one structure gradually separate into increasingly independent relationships. Momentum remains present, but the various Momentum are no longer organized together in the same way to maintain the same integrated configuration. Heat continues to move outward, chemical reactions proceed toward different equilibria, cells undergo changes at different rates under different conditions, and microorganisms form new relationships through pathways that had previously been constrained. Death should therefore be understood less as the instant when activity ends than as the process through which an integrated Momentum Structure is progressively dismantled and reorganized into multiple other Momentum Structures.

This transition can begin even while the outward form of the organism remains recognizable, because what matters is not the immediate disappearance of its physical outline but a change in the **Effective Boundary** that had maintained the organism as one living Island. An Effective Boundary is not merely a wall physically separating inside from outside. It is a structural boundary through which internal and external relationships are selectively organized and regulated. While an organism is alive, matter, energy, and information continually cross this boundary under selective conditions. Some external structures are admitted, others are excluded or constrained, while Momentum organized internally is transmitted back into external relationships. Through this continuing selective organization, the organism maintains itself as an integrated Momentum Structure.

When death begins, it is this effectiveness that changes. The skin may remain physically present for some time, the organs may remain in approximately the same locations, and the outward appearance of the body need not immediately collapse. Yet the functions that had organized inside and outside as parts of one living relationship no longer operate in the same way. While the organism was alive, its Effective Boundary sustained internal Differences while continuously regulating selective exchange with the outside; damaged structures could be repaired, local changes could be readjusted within the stability of the whole, and matter, energy, and information could be circulated and reorganized in ways that contributed to maintaining the living Island.

As the integrated Momentum Structure begins to collapse, this organization weakens with it. External microorganisms can participate more readily in internal relationships, microorganisms already present within the body can expand into regions from which they had previously been constrained, chemical conditions begin shifting along paths different from those maintained during life, and cells can no longer regulate their activity in accordance with the integrated requirements of the organism as a whole. Tissues gradually break down, while matter and Momentum that had been maintained within one living Island enter new relational networks. The important point is that the boundary need not physically disappear in order to cease being structurally effective. The body may retain its visible outline immediately after death, but that outline no longer organizes inside and outside as the Effective Boundary of the same living Island.

The transition is also not completed uniformly across every scale. A living organism is not one simple structure but a **Nested Structure** formed through the close integration of numerous subordinate Islands. A cell is an Island. Tissues are larger Islands formed through relations among cells. Organs integrate multiple tissues, and the organism as a whole forms a still larger Island through the close organization of its organs and other structures. A living organism is, in this sense, an Island composed of Islands, and its death cannot therefore occur at exactly the same moment at every level.

At the level of the organism, life may be judged to have ended while biological activity continues for some time at the cellular level. At the molecular level, chemical relationships persist, while at the level of microorganisms new structures may actually begin to organize more actively. Death is therefore not one event occurring simultaneously at every scale but a structural transition proceeding differently across different scales. This is also why medicine does not rely on one simple phenomenon as an exhaustive description of death. Cardiac function can sometimes be restored after it has stopped, respiration can be artificially maintained, some functions of the nervous system can disappear while others remain for a period, and organs and cells can retain viability after the organism as a whole has been declared dead.

These differences do not arise merely because medicine has not yet found a perfect definition. More fundamentally, the organism itself is a multilayered Momentum Structure, a complex Island in which numerous subordinate structures are bound together in different ways. The disintegration of that integrated structure can consequently proceed gradually across several levels rather than occurring as one universal instant. What death means therefore also depends on which Island has been selected as the object of observation. If the organism is treated as one Island, there is the death of the organism; if the cell is treated as an Island, the relevant question becomes the viability of the cell; from the perspective of a larger ecosystem, a broader relational network may continue even as one individual disappears. As the observational scale changes, so does the range of the structural transition described as death.

This does not reduce death to a choice made by the observer. Death is a real structural change. The integration that maintained one living Island actually collapses, selective responses to the outside disappear, self-maintaining processes can no longer be sustained, and the Effective Boundary loses its function of organizing one living structure. These changes occur regardless of how an observer describes them. What can depend on observational scale is which structure is selected as one Island and at which level the transition is called death. Death is therefore real without being absolute: neither an illusion nor one universal event occurring simultaneously at every level, but an actual structural transition in which a particular Island can no longer continue as the same integrated living structure.

Death thus has both reality and relativity. What is real is the collapse of an integrated structure; what is relative is which structure we select as one Island and identify that collapse as its death. This distinction also changes the meaning of the familiar expression *loss of life*. At death, no independent substance called life leaves the body. What disappears is not a material called life but the integrated organization that had allowed an Island to exist as a living structure. Momentum paths that had been tightly interconnected within one Effective Boundary are no longer organized in the same way. Some pathways cease to remain effective in their previous form, some continue independently, and others enter new relationships with surrounding Islands. A living Island does not turn into nothingness; its constituent structures are reorganized within other relationships.

The same structural principle need not be restricted to the level of an individual organism. A species can disappear, an ecosystem can collapse, and a civilization can lose the form by which it had been recognized through history. In such cases, not every constituent disappears simultaneously. Individuals may disperse, matter may enter other structures, and some relationships may continue, even though the relational network that integrated them into one distinguishable configuration no longer persists. In this broader structural sense, the loss of integration by which a particular configuration maintained itself can occur in every kind of Island. A star can enter a new configuration when its internal relationships can no longer sustain the structure it previously maintained; mountains lose earlier configurations through uplift and erosion; storms dissipate, rivers change course, and machines lose their functional organization. We do not ordinarily call these changes death, but the structural principle—that an Island can lose the organization through which it had maintained a distinguishable configuration and be reorganized into other relationships—is not exclusive to living systems.

The death of a living organism nevertheless feels profoundly different because a living Island maintains an exceptionally high degree of Dynamicity and selective organization. When that integration collapses, what is lost is not merely a particular arrangement of matter but a distinctive structural identity. Understanding death as structural transition does not diminish the reality of personal loss or grief. A person is not simply an aggregation of matter but a unique Momentum Structure in which body and nervous system, memory and experience, and countless relationships with other people have been integrated into one continuing configuration. When that configuration can no longer be sustained, the loss is real. The continued existence of its constituent materials does not mean that the Island that was that person continues to exist, just as the survival of paper and ink would not by itself preserve the content and meaning of a book whose organization had been destroyed.

What is lost in death, then, is not matter but the structure of relationships that organized that matter as one living Island. Death must therefore be understood through continuity and discontinuity at once. In terms of continuity, Momentum in the Universe does not stop; Differences continue to participate in new relationships, Momentum continues to become effective through changing paths, and Equalization continues through the relational networks of countless Islands. In terms of discontinuity, however, the integrated Momentum Structure that maintained one living Island can no longer continue as that same structure. Death is the point at which these two facts meet: the wider relational processes of the Universe continue, while a particular Island no longer persists as the Island it had been.

Life and death are therefore not absolute opposites dividing existence from nonexistence. Life is a structural state in which an integrated Momentum Structure maintains an Effective Boundary and organizes a high degree of Dynamicity; death is a structural transition in which that integration can no longer be sustained and the constituent structures enter new relational networks. Both occur within the same continuous relational Universe. The more consequential question is therefore no longer simply **"What are life and death?"** but how a living Island, operating within the same relational principles as every other Island, behaves differently and participates in the resolution of Momentum in its own distinctive way.

7 Beyond Life and Death

Life and death no longer occupy the positions in which we first placed them. They are not an absolute boundary dividing the Universe into two different domains of existence. Life and death are not distinguished by whether matter exists, nor by whether energy exists, nor by whether Momentum exists. Neither are they separated by whether change occurs. Everything in the Universe changes. Everything we recognize as a distinct entity is an Island formed through relationships among smaller Islands; every Island maintains some degree of internal binding, enters into relationships with surrounding Islands, and participates in the organization and resolution of Momentum. A stone in a field, a star in space, a dormant seed, a growing tree, an animal, and a dead organism all remain within the same relational Universe. What distinguishes them is not that some exist while others do not, but how Momentum is organized, what kind of structure that organization forms, and how its resolution proceeds.

From this perspective, the argument developed through this chapter can be condensed into three propositions. First, every distinguishable entity is a **Momentum Island**. Nothing we recognize as an Island exists completely detached from relationships; it is distinguishable because its internal relations are organized strongly enough, relative to surrounding relations, to maintain a persistent configuration. Second, every Island occupies a region within a continuous spectrum of **Dynamicity**. The organization of internal relations, the density and integration of Momentum paths, and the selectivity through which they become effective differ from one Island to another, but these differences occur across continuous structural variation and do not require an absolute ontological discontinuity between life and nonlife. Third, life and death are structural regions identified by an observer within that spectrum. Life refers to a region in which Dynamicity attains particularly high density and organization, while death refers to a structural state in which such integration can no longer be maintained and the configuration passes into other relationships.

These propositions are not intended to abolish the distinction between life and death. They place that distinction more precisely. Life and death remain important biological categories, as well as indispensable medical, ethical, and social ones. A living body and a dead body are not the same structure. The collapse of the integrated Momentum Structure that maintained an organism is real, and the loss associated with that collapse is equally real. Yet this does not require the Universe itself to be divided into two separate ontological domains called life and nonlife. The Universe remains one continuous relational process, within which life and death appear as different structural states.

A different question now emerges. If every Island participates in Momentum and every Island participates in the processes of Equalization, why do living Islands appear so different from stones, stars, rivers, flames, and other structures? To answer that they are different because they possess life would explain nothing; it would merely define life by referring back to life. Nor can the difference be explained by the amount of activity involved. A star releases vastly more energy than an organism. A storm continuously reorganizes structures across an immense region, a river reshapes terrain over long durations while establishing new relationships, and a flame rapidly transforms surrounding material in ways that create further changes. All of these display substantial Dynamicity, yet we do not call them living organisms. Conversely, living organisms are not always visibly active. A dormant seed may appear almost unchanged, and a hibernating animal may remain outwardly almost static, yet we continue to recognize both as living.

The distinction between life and nonlife therefore cannot be reduced to more or less activity, to the magnitude of energy or Momentum, or to the speed at which change becomes visible. DISTANCE locates the difference instead in **how Momentum is organized**. Every Island participates in Momentum, but not every Island organizes its Momentum relationships in the same way. Some maintain Momentum through a relatively limited set of relationships, while others organize many different relations simultaneously. Some respond to changes in their surroundings largely within pathways constrained by their existing structure, whereas others continually reorganize their relations with the outside and, through that reorganization, alter the relational network in which they themselves participate.

This is where the distinctive character of a living Island begins to become visible. Life does not possess a new principle unavailable to the rest of the Universe. Rather, relational principles already present across Islands become organized and expressed within living structures at a much higher degree of integration. The more fundamental question is consequently no longer simply, **“What distinguishes life from nonlife?”** It is how a living Island, while remaining within the same relational principles, becomes organized and operates differently from other Islands. Following that question leads beyond classification and toward the particular way living Islands participate in Momentum Resolution.

The distinction is not merely that living Islands contain more processes. Stones possess internal relations. Stars maintain immense networks of interacting physical processes. Rivers, atmospheres, and flames continuously interact with their surroundings. In many such structures, however, the Momentum paths that become effective remain comparatively constrained by the existing configuration and surrounding conditions. The structure changes, but the extent to which those changes reorganize the conditions for forming further relations is relatively limited. Within living Islands, by contrast, different Momentum paths continuously affect one another. A Difference arriving from outside can alter several internal paths at once; changes within those paths modify the conditions under which other paths become effective, and one relational change can alter the possibility of forming another. The configuration consequently continues to readjust the organization through which later Momentum becomes effective.

What matters here is not simply the number of paths but their **connectivity**. Hundreds of processes existing simultaneously but largely independently do not necessarily produce higher Dynamicity than a smaller number of paths that are tightly interconnected and continually reorganize one another. Dynamicity is therefore not identical with complexity. A structure containing many components does not automatically possess high Dynamicity, whereas a structure with fewer components can display substantial Dynamicity if changes among them remain tightly coupled and repeatedly reconstruct the configuration as a whole.

The distinctive character of life consequently lies neither in possessing more matter, using more energy, nor moving more rapidly. It lies in the degree to which numerous Momentum paths are connected within an Effective Boundary and continually modify the conditions under which one another are organized. A living organism is therefore not merely a structure that changes; it is a structure in which **change itself becomes organized through an interconnected relational network**. Yet even this organization must not be interpreted as a principle invented exclusively for life. It is a more highly integrated expression of relational processes already present throughout the Universe.

Consider how this difference appears in the relationship between inside and outside. A stone resolves Momentum within the relations permitted by its structure and surroundings. Pressure can produce fractures, water and air participate in weathering, and exchanges of heat gradually alter its configuration. These are genuine changes, and they remain part of the broader processes of Equalization. A living Island, however, does more than respond through a relatively fixed set of available relations. Its internal organization changes according to its relations with the outside, and the changed internal organization in turn alters the way subsequent external relations become effective. Inside and outside therefore do not form a simple one-way sequence; they constitute an interconnected network in which each can continually modify the conditions of the other.

A plant changes the direction of root growth according to surrounding conditions. An animal moves and thereby enters environments in which relationships unavailable at its previous location can become effective. A cell responds to different external Differences by organizing different combinations of internal pathways. The significance of these examples is not that life alone possesses some mysterious capacity for response. It is that, in living Islands, relations between inside and outside are repeatedly **reorganized**. One change modifies the conditions for another, and that modification affects the way later relations can form. A living Island does not persist by preserving its present relations unchanged; it remains a living structure by continually reorganizing them.

Living Islands therefore remain within the same relational principles as stones and stars while displaying a markedly different form of their expression. Their internal organization also changes the way they enter the wider network of other Islands. A living Island maintains relations already organized within its boundary while continually forming new relations with many external Islands. Structures that previously had no direct relation may thereby become connected within one network, while existing relations may be reorganized in ways that make different Momentum paths effective.

Nothing in this process requires a purpose imposed upon the Universe. A plant does not extend its roots toward water in order to assist cosmic Equalization. An animal does not search for food, nor does a microorganism carry out metabolism, because it intends to resolve the Momentum of the Universe more efficiently. Each Island forms relations through the conditions under which its own structure persists, and those relations alter the conditions under which further relations can become effective. The important point is therefore not the intention of an individual act but the structural consequence that, through such acts, the relational network itself continues to change.

A living Island can incorporate external structures into its own internal relations and can also extend structures formed internally back into external relations. In doing so, one Difference becomes connected with another, and one Momentum path becomes coupled with other Momentum paths, allowing a greater variety of Momentum Resolution paths to become structurally available within the wider network. This **expansion** does not mean that living systems create Momentum that did not previously exist. Momentum is always present. What expands is the structural range within which existing Difference and Momentum can enter new relations and become effective through newly organized paths.

A living Island, in this sense, is not a generator that produces Momentum from nothing. It is a structure that can enrich the network through which Momentum is organized, transmitted, and resolved. Its distinctive Dynamicity does not remove it from the rest of the Universe; it connects more relations within that Universe and repeatedly changes the conditions under which those relations can become effective. The significance of life thus lies not only in maintaining a highly integrated internal configuration but also in the way that configuration can broaden and reorganize the relational field around it.

This brings the argument beyond the original opposition between life and death. The central issue is no longer where an absolute line should be drawn between them. It is how living Islands form the dense relational networks through which their internal organization is maintained while their relationships with other Islands continue to expand, and how those expanding relations diversify the paths through which Momentum can be resolved. Life is not where Momentum begins. Nor is death where Momentum ends. The distinctive problem of life begins with the extraordinary way in which already-present Momentum becomes interconnected and reorganized through living Islands, creating conditions under which still more relations can become effective.

What remains to be understood is the structural mechanism by which living Islands accomplish this: how they exchange with their surroundings, incorporate other structures, reorganize what enters their boundaries, form relations with other living Islands, and continually extend the network of available Momentum Resolution paths. That is where the continuous spectrum of Dynamicity begins to reveal not merely what life **is**, but what living Islands **do within the wider relational Universe**.